Two-way tables

1 Members of a gym were surveyed about what kind of training they do. Each responder only did one kind of training and the results are given in the following table:

	Cardio	Weight
Male	12	26
Female	44	18

a Find the total number of gym members surveyed.

b Find the number of members who do weight training.

c Find the percentage of the gym members who do weight training.

2 A survey on smoking was conducted and the results are given in the table below:

	Smokers	Nonsmokers
Men	35	79
Women	56	139

a Find the number of the people surveyed who were smokers.

b Find the percentage of women who were nonsmokers. Round your answer to two decimal places.

c Find the percentage of nonsmokers who were women. Round your answer to two decimal places.

3 The departure timing of flights out of an airport for the months of May and June are described in the given table:

	Departed on time	Delayed
May	144	21
June	116	40

a Find the percentage of the flights in June that were delayed. Round your answer to one decimal place.

b Find the percentage of the flights in June that were on time. Round your answer to one decimal place.

c What fraction of the total number of flights during the two months were ones that departed on time in May?

4 Katrina tossed three coins. She displayed the results in the given two-way table:

	Heads	Tails
Coin 1	19	12
Coin 2	13	14
Coin 3	15	13

a How many times did Katrina toss heads?

b How many more times did coin 2 land on tails than coin 1?

c How many times did Katrina toss the coins in total?

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5 Two motels promoted packages for one night and two night stays. The following table records the results:

	One night stay	Two night stays
Motel X	463	266
Motel Y	804	817

a Find the percentage of the stays that are two night stays. Round your answer to two decimal places.

b What fraction of the two night stays was in Motel X?

c Find the percentage of the total accommodation that is in Motel X. Round your answer to two decimal places.

6 Michael rolled two dice. He recorded how many times each die landed on each score in the given two-way table:

Number rolled	Die 1	Die 2
1	17	13
2	9	16
3	12	7
4	10	4
5	9	6
6	7	8

a How many times did he roll a five?

b How many times did Michael roll die 2 ?

c With which die did he roll more fours?

7 In a biology study, the appearance of an animal when carrying a certain gene is recorded. Assume that all animals must carry Gene A or Gene B but not both. If, out of all the animals, the proportion carrying Gene A and having light fur is $\frac{4}{19}$, find the value of x.

	Gene A	Gene B
Light fur	36	37
Dark fur	28	x

8 500 people were asked whether they own a car or a motorcycle. The survey had the following results:

- 15% of people curveyed own a motorcycle.
- 70% of people surveyed own a car.
- 80 people surveyed do not own a car or a motorcycle.

Construct a two-way table to represent these results.

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9 When people were asked if they found certain kinds of food addictive, the relative frequency of their responses was recorded:

	Addicted to caffeine	Not addicted to caffeine
Addicted to sugar	0.3	0.3
Not addicted to sugar	0.3	0.1

a What proportion of responders find at least one of the two products addictive?

b If 90 people took part in the survey, how many found at least one of the products not addictive?

10 Thirty six students were asked whether or not they were allergic to nuts and dairy. The two way table is provided below:

	Allergic to nuts	Not allergic to nuts
Allergic to dairy	10	6
Not allergic to dairy	6	14

a How many students are allergic to nuts?

b How many students are allergic to nuts or dairy, or both?

c How many students are allergic to at most one of the two things?

11 In a study, some people were asked how many times they lie in a day. 20 responders said they lie at least once a day, of which 5 were children. 15 children said they never lie, and 15 adults said they never lie.

a Complete the given table.

b What proportion of responders said they never lie?

c What proportion of adults said they lie at least once per day?

	0 times	1 or more times
Children		
Adults		

12 A survey was given to mall-goers to determine their favorite pie flavor. The results are shown in the table:

	Apple	Blueberry	Pecan
Children	124	104	70
Adults	124	42	135

To the nearest percent, what percent of the adults chose blueberry?

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13 The following table shows the number of trains arriving either on time or late at a particular station:

Day	Arriving on time	Arriving late
Monday	23	38
Tuesday	12	25
Wednesday	22	29
Thursday	30	27
Friday	11	14
Saturday	15	11
Sunday	14	10

a How many trains were late on Friday?

b How many trains passed through the station on Wednesday?

c How many trains were on time throughout the entire week?

d What fraction of trains were on time over the whole week?

14 A total of 150 tennis players were asked whether they would support equal prize money for the women's and men's draw.

	Support	Do not support
Males		35
Females	66	12

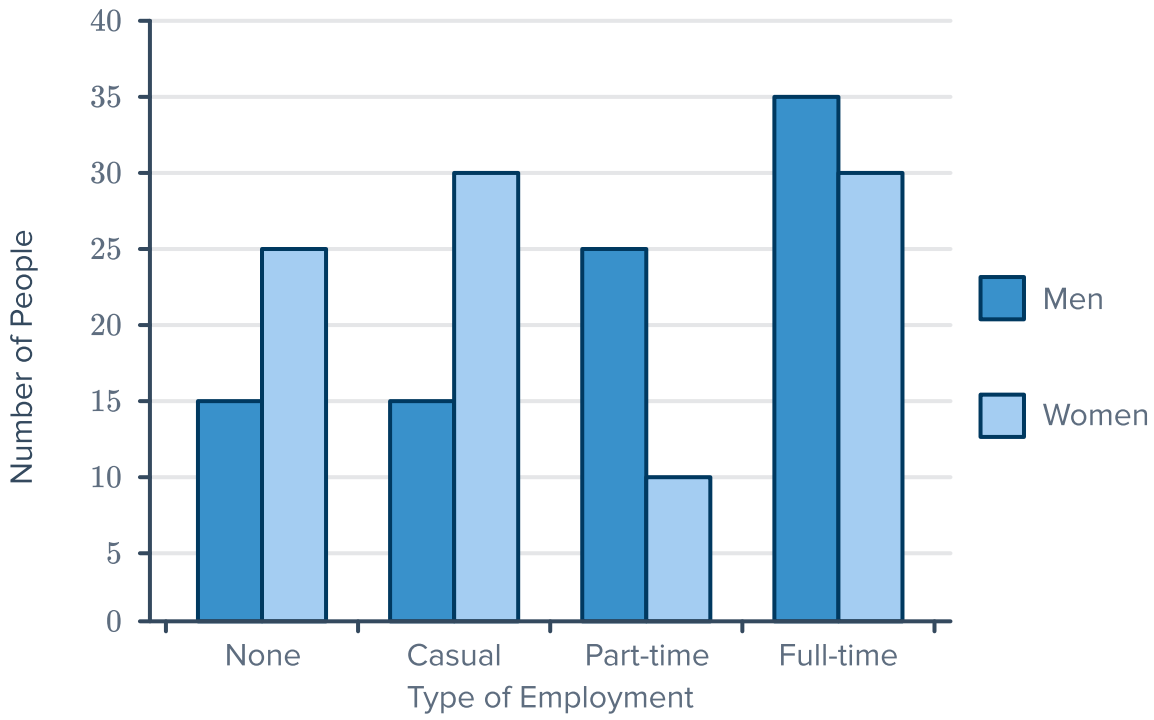
a Find the missing value in the table.

b How many more players are there in support of equal prize money than those against it?

c Find the percentage of the male tennis players who support equal prize money. Round your answer to one decimal place.

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15 Yuri surveyed a group of people about the type of jobs they had. He recorded the data in the following graph:



a Complete the following two way table with the information:

	No job	Casual	Part-time	Full-time
Men				
Women				

b How many people are unemployed?

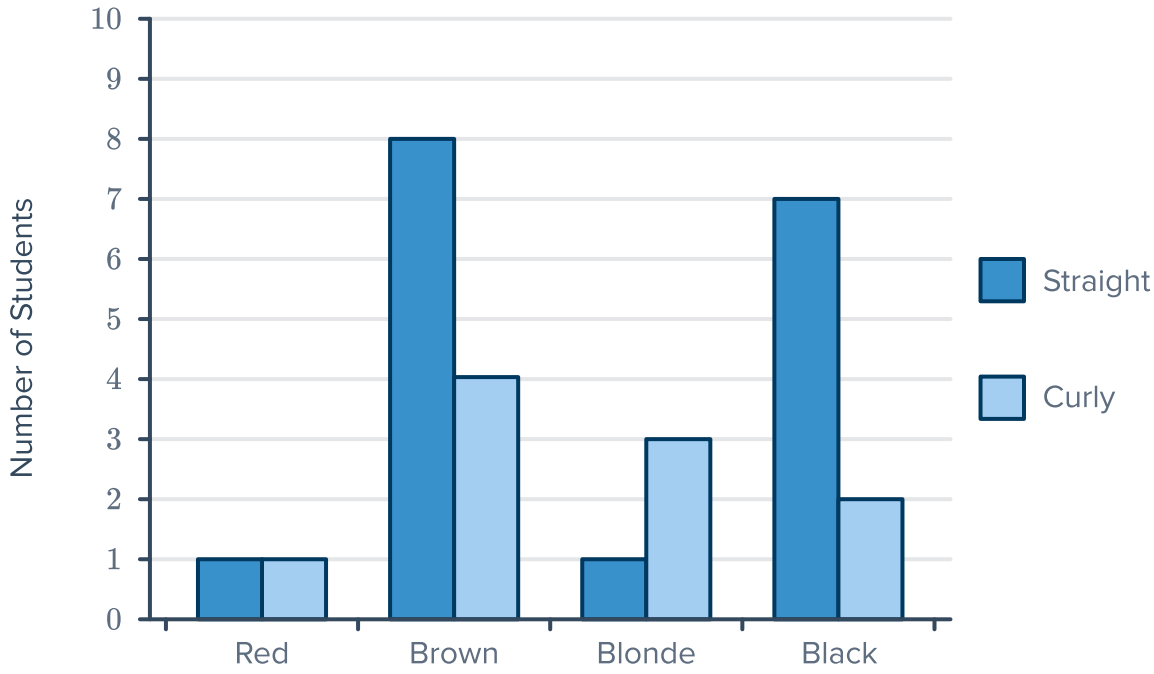
c What percentage of men are employed full-time? Round your answer to one decimal place.

d What fraction of people employed casually are male?

e If 111 people from the group are randomly selected, how many would you expect to be employed part-time?

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16 Miss Merryweather recorded the hair type and color of her students in the following graph:



a Complete the following two way table with the information:

	Red	Brown	Blonde	Black
Straight				
Curly				

b Find the number of students in the class that were surveyed.

c What fraction of students have black hair?

d If 15 students from the class are randomly selected, how many would you expect to have brown hair? Round your answer correct to the nearest whole number.

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